

## **Scope Of Work**

The agency shall develop and deploy an AI-based solution capable of:

- The IPD bills are to be uploaded by the hospitals at the NTPC portal. Integrating these bills with existing system i.e. Hospital Information System through APIs to fetch the invoices/data. Extract data from scanned/digital IPD bills and discharge summaries, reports etc using advanced OCR.
- Standardizing, validating and processing the extracted data.
- Matching invoice line items with Schedule of Charges (SOC), MoUs, and Medical Rule Sets.
- Detecting anomalies, frauds, and incorrect treatments.
- Exporting validated data to Excel/JSON/XML or SAP-compatible format.
- Generating analytical reports and dashboards.
- System should handle a minimum of 100 +/- 20% invoices/day without degradation. Annually approx. 50,000 invoices are being processed. At present approx. 260 hospitals are empanelled with NTPC at present. This is variable and can increase in future
- Concurrent processing by multiple users functions seamlessly
- Except for desktops/PCs. all other necessary hardware shall be provided by vendor.

| Module / Section                    | Feature                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Document Ingestion & Pre-processing | Multi-form Support          | Accepts and parses PDFs, scanned images, (pdf/images of prescription, diagnostic reports, imagery, etc) and EMR files. Offers drag-and-drop or API-based ingestion.                                                                                                                                                                                                                                                             |
|                                     | Metadata Extraction         | Captures file metadata such as patient ID, claim number, hospital name, and admission dates.                                                                                                                                                                                                                                                                                                                                    |
| OCR & NLP Engine                    | AI-Powered OCR              | Converts scanned documents into machine-readable text using advanced OCR models. Accuracy of OCR should be more than 95%                                                                                                                                                                                                                                                                                                        |
|                                     | NLP-based Structuring       | Extracts key medical data points—diagnosis, procedure codes, drug names, doctor notes, etc.—using named entity recognition (NER).                                                                                                                                                                                                                                                                                               |
|                                     | Data storage and conversion | <p>Convert extracted data into a standardized format compatible with SAP tables and Excel. Ensure consistency across diverse bill formats from multiple hospitals.</p> <p>The data shall include IPD invoices, Schedule of Charges (SoCs), and Memorandums of Understanding (MoUs) with all empanelled hospitals. The system shall be capable of incorporating any addendums or subsequent changes made to these documents.</p> |

|                                   |                                       |                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anomaly & Abuse Detection Engine  | Clinical Rule Engine                  | The tool should provide rich library of clinical rules which Detects billing patterns such as phantom billing, duplicate claims, unbundling of procedures, over-medication, and treatment non-compliance. Flag inconsistencies between diagnosis, procedures and prescribed medications using clinical rule sets and machine learning models. |
|                                   | Document Validation                   | Flags missing documents, inconsistent patient details, or potential for forged/mismatched entries.                                                                                                                                                                                                                                            |
|                                   | Tailored Trigger Logic                | Provides customizable rules for flagging high-risk claims, unusually long stays, or repeated procedures.                                                                                                                                                                                                                                      |
|                                   | Data validations                      | Validate billing amounts and treatments against dynamic MOUs, SOCs, and NCA rule sets.                                                                                                                                                                                                                                                        |
|                                   |                                       | Use statistical data, machine learning, artificial intelligence, clinical, and rule based classification mechanisms for categorization of claims, fraud detections using both structured and unstructured data.                                                                                                                               |
| Data processing                   | Workflows and dashboards              | Develop a user-friendly dashboard for medical officers and finance team to monitor anomalies, review reports, and manage reverification workflows.                                                                                                                                                                                            |
| Historical Pattern Correlation    | Medical History Integration           | Maps new claims against previous records and flags unusual frequencies or progression patterns in historical data.                                                                                                                                                                                                                            |
|                                   | Behavioural Insights                  | Uses predictive logic to identify potentially avoidable admissions or chronic overuse patterns.                                                                                                                                                                                                                                               |
| Compliance Validation             | Standard Treatment Protocol Alignment | Validates diagnoses and treatment paths against standard treatment guidelines set by MOHFW, NHSRC, ICMR, NCDC, <b>The Clinical Establishments, AIIMS Guidelines, CDSCO</b> or other regulatory bodies.                                                                                                                                        |
| Adaptive Learning Module          | Feedback Loop                         | Allows reviewers to approve or override AI flags; these inputs are used to re-train the model periodically.                                                                                                                                                                                                                                   |
|                                   | Human-in-the-Loop                     | Incorporates expert review to minimize false positives/negatives through active learning.                                                                                                                                                                                                                                                     |
| Performance & Reporting Dashboard | Claims Audit View                     | Enables filtering and detailed drill-down review of high-risk claims, flagged cases, and rejected claims.                                                                                                                                                                                                                                     |

|                          |                       |                                                                                                                                                                       |
|--------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Visual Analytics         | Dashboards            | Provides comprehensive visualizations of admission trends, hospital performance, fraud patterns, readmission rates, and policy utilization.                           |
| Export & Traceability    | Export & Audit Trails | Enables export of flagged cases, generates audit trails, and maintains logs to support compliance and legal review processes.                                         |
| Integration & Deployment | NTPC Integration      | Integrates seamlessly with NTPC's internal portals and claim management systems via APIs.                                                                             |
|                          | Deployment Options    | *Fully scalable cloud-based system with on-premises integration with SAP, NTPC IPD portal and Excel exports(if any).<br><b>(*Hyperscale should be Meity approved)</b> |

#### Support Requirements:

- Collaborate with the NTPC team to incorporate SOC's, MOUs, and medico-financial logic.
- Provide comprehensive training and user manuals for NTPC staff.
- Offer ongoing support for system updates, including integration of new MOUs and SOC's.
- Conduct quarterly performance reviews and incorporate user feedback.
- Service Availability

| Service Component           | Availability | Commitment | Measurement Period |
|-----------------------------|--------------|------------|--------------------|
| Application uptime          | 99.9%        |            | Monthly            |
| Cloud infrastructure uptime | 99.9%        |            | Monthly            |
| API availability            | 99.9%        |            | Monthly            |

#### • Performance Matrix

| Metric                         | Target                       | Measurement              |
|--------------------------------|------------------------------|--------------------------|
| Invoice processing speed       | ≥ 500 Pages/hour             | Avg. per batch           |
| OCR accuracy (text extraction) | ≥ 95% for standard templates | Random monthly sampling  |
| Data validation accuracy       | ≥ 98%                        | Based on matched records |

| Metric                 | Target                              | Measurement          |
|------------------------|-------------------------------------|----------------------|
| Response time (UI/API) | ≤ 2 seconds for 95% of transactions | Synthetic monitoring |

## Documentation

- Submit a detailed project plan with timelines, milestones, and resource allocation.
- Provide technical architecture diagrams, data flow models, and API documentation.
- Include a risk mitigation plan addressing challenges (e.g., OCR inaccuracies, evolving medical rule sets).
- Deliver system maintenance and troubleshooting guides.

## Support & Maintenance

Minimum 12 months of post-deployment support, including updates to accommodate changes in SOC/MoUs and onboarding of new empanelled hospitals.

## Terms and Conditions

- **Contract Duration:** The contract shall be valid for a period of **one year** post successful implementation. The **implementation period shall be four (04) months** from the date of award of contract.

**Payment Terms:** Payments shall be milestone-based as follows for Implementation and deployment Cost:

- 20% upon successful system deployment and integration
- 40% upon completion of acceptance testing and Go-Live and submission of documentation.
- 40% after 45 days of Go-live

No advance payments shall be made.

Payment shall be made through monthly RAB for remaining BOQ items

- **Support and Maintenance:** The bidder must provide 24/7 technical support for the contract duration and a minimum of 1 years post-implementation. This support must encompass all issues, including but not limited to system downtime, software bugs, performance degradation, any minor changes, modifications, or new requirements that may arise during the support period, data integrity failures, AI model inaccuracies, integration errors, cybersecurity incidents, hardware compatibility problems, user access issues and any other functional or operational disruptions affecting the AI-driven healthcare solution.
- **EMD: as per GCC**

- **LD Clause: as per GCC**
- **Termination Clause:** The contract may be terminated with 60 days' notice for non-performance or breach of terms.
- **Performance Guarantee:** A bank guarantee of 5% of the contract value must be submitted upon award valid for entire contract duration plus three months.
- **Intellectual Property Rights:** IP right shall be applicable as per the law of land
- Dedicated team with expertise in AI/ML, OCR, data extraction, and healthcare domain knowledge.
- All documents are encrypted at rest and in transit (TLS 1.2+).
- **Final Acceptance Criteria:** All functional modules pass internal and user test cases with at least 90% precision/recall in fraud detection with defined benchmarks. System is approved for production deployment by NTPC Audit/Medical Team.

Documentation, training manuals, and support SLAs are delivered.

- **Confidentiality & Data Security**  
Agency must ensure full compliance with data protection laws and institutional policies. No data is to be shared or used outside the agreed scope.  
Bidders must sign a Non-Disclosure Agreement (NDA) to protect sensitive hospital and employee data.
- The Agency must ensure strict compliance with applicable Indian data protection laws, including the Information Technology Act, 2000 (as amended), relevant rules under it, and any other government-mandated standards such as the Digital Information Security in Healthcare Act (DISHA), once in force and MEITY guidelines/regulations. All necessary safeguards to protect health related data under this project to be ensured.

No data, in any form, shall be shared, transmitted, or used beyond the agreed project scope without prior written approval from the competent authority.

Successful Bidder shall execute a Non-Disclosure Agreement (NDA) to safeguard sensitive hospital, patient, and employee information.

The vendor shall adopt reasonable security practices such as encryption, role-based access control, secure authentication, audit trails, and regular vulnerability assessments to protect all personal and health-related data handled under this project, in compliance with IT Security policies and applicable Indian legal requirements.

The evaluation of the tender shall be carried out on a lump sum basis.

### **Bill of Measurement**

| <b>Sl</b> | <b>Item description</b>               | <b>Unit</b>   | <b>Quantity</b> |
|-----------|---------------------------------------|---------------|-----------------|
| <b>1</b>  | Implementation and deployment Cost    | LPS           | 1               |
| <b>2</b>  | Post-implementation Support (1 Years) | Months        | 12              |
| <b>3</b>  | <b>Variable cost per page *</b>       | <b>Page</b>   | <b>1000000</b>  |
| <b>4</b>  | <b>Fixed cost (monthly)</b>           | <b>months</b> | <b>12</b>       |

**\*Variable cost per page is applicable on actual page processed. Estimated annual volume: 10,00,000 pages (50,000 invoice X 20 pages avg.). billed monthly as per actual.**

**One time implementation/deployment cost is inclusive of the following:**

- Implementation Services Fees, which shall cover cost for installation, setup including all hardware, and initial configuration.
- AI software license fees (includes NLP, OCR, ML Modules)
- System Integration Includes API development and integration with existing systems such as SAP or Hospital Information Systems (HIS).
- Data migration and cleansing for historical billing and medical data
- Custom reports and dashboards for analytics and monitoring
- User training for administrators and end users.
- Cost for Cloud infrastructure hosting.
- Documentation for technical and end-User manuals